Attention Mechanisms in Deep Learning

Introduction

Attention is a mechanism that allows a model to dynamically focus on different parts of its input when producing each element of its output. It was originally introduced to improve neural machine translation and has since become foundational in modern deep learning.

Before attention, sequence-to-sequence models compressed the entire input into a single fixed-length context vector, creating an information bottleneck. Attention solves this by allowing the decoder to look back at all encoder states.

Bahdanau Attention

Bahdanau et al. (2015) introduced the first widely-used attention mechanism for machine translation. At each decoding step, an alignment score is computed between the current decoder state and every encoder hidden state.

These scores are passed through a softmax to produce attention weights, which are then used to compute a weighted sum of encoder states, called the context vector. The context vector changes at every step, letting the decoder focus on different source words.

Scaled Dot-Product Attention

The Transformer architecture introduced a simpler and more parallelizable form: scaled dot-product attention. Given queries Q, keys K, and values V (all matrices), attention is computed as: $\text{Attention}(Q, K, V) = \text{softmax}(Q * K^T / \text{sqrt}(d_k)) * V$.

The scaling factor $\text{sqrt}(d_k)$ prevents the dot products from growing too large in high dimensions, which would push the softmax into regions with very small gradients.

Multi-Head Attention

Rather than performing a single attention function, Multi-Head Attention runs h attention operations in parallel, each with its own learned linear projections of Q, K, and V. The outputs are concatenated and projected again.

This allows the model to jointly attend to information from different representation subspaces at different positions, capturing diverse types of relationships simultaneously.

Self-Attention

In self-attention, queries, keys, and values all come from the same sequence. Each token attends to all other tokens in the sequence, allowing the model to capture long-range dependencies without the limitations of recurrent models.

Self-attention is the key building block of Transformers and is responsible for their ability to model complex dependencies regardless of distance in the sequence.

Cross-Attention

Cross-attention is used in encoder-decoder Transformers (e.g., for translation or summarization). The decoder queries come from the decoder's own representations, while keys and values come from the encoder output.

This allows the decoder to selectively attend to relevant parts of the encoded input when generating each output token.

Applications Beyond NLP

Vision Transformers (ViT) apply self-attention to patches of images, treating each patch as a token. This has achieved state-of-the-art results on image classification.

Cross-modal attention connects different modalities, such as in CLIP or DALL-E, where text and image representations interact.

Attention is now used in protein structure prediction (AlphaFold), speech recognition, video understanding, and many other domains.